

Validation-Swap Distributions

Diagnosing Machine-Scored Outcomes in Experiments

August 2026

Abstract

Researchers increasingly use machine predictions or large language model scores as experimental outcomes, then obtain human labels for a smaller validation sample. Average prediction accuracy does not show whether scoring error distorts the treatment contrast. We study the distribution of treatment-effect estimates over fixed-size sets of clusters whose machine scores are replaced by human labels. For a linear treatment contrast, the mean raw-swap path is mechanically determined by its endpoints. Its dispersion and tails contain the information about whether distortion is diffuse, concentrated, or offsetting. Scaling the swap correction by its sampling probability yields a standard design-based difference estimator, not a new correction method. The swap distribution instead shows how the corrected estimate varies with the realized validation sample. In an exactly enumerated experiment, two scoring systems have the same RMSE, R^2 , proxy effect, gold effect, and mean swap path, yet a five-cluster swap has no effect 55.3% of the time when scoring error is concentrated. The diagnostic is conditional on the observed gold-label pool. Population inference still requires a probability validation design or a transport model.

Researchers can score every response in an experiment with a classifier, a computer-vision model, or a large language model while paying human coders to score only a fraction. The cheap score may have excellent test-set accuracy and still distort the treatment effect. The relevant error is not average scoring error. It is the difference in scoring error between treatment arms.

Suppose an experiment tests whether a message changes the hostility expressed in an open-ended response. A language model scores hostility for all respondents, and human coders score a random subset. The model can overstate hostility by two hundredths in one arm and understate it by two hundredths in the other. Its mean error is zero, but the estimated treatment effect is wrong by four hundredths. Aggregate accuracy conceals this error because accuracy and the treatment contrast weight observations differently.

Existing methods already correct downstream inference with a small set of gold labels. Post-prediction inference, design-based supervised learning, and prediction-powered inference combine predictions on many observations with gold-label residuals on a smaller probability sample [6, 5, 2]. Two-phase validation designs study how to select the expensive measurements efficiently [1]. Work on text as data explains why a learned outcome requires separate measurement and causal design choices [4]. These methods establish the estimator and its uncertainty.

This paper asks a different question: how much does the answer depend on which clusters receive the gold labels? We define the fixed-budget random-swap distribution of the treatment-effect estimate.

Its mean is often uninformative. Its spread shows whether a validation budget produces a stable correction or a lottery. Its tails show how often the estimate moves by more than a prespecified substantive amount. The same construction applies to people, documents, sites, or other sampling units, but the swap unit must match the validation design and the dependence structure.

The contribution is deliberately narrow. The design-scaled estimator below is a special case of established difference and prediction-powered estimators. The finite-population variance formula is standard sampling theory. We use these results to define an estimand-specific validation diagnostic and to show why an average swap path can hide the feature an audit is meant to find.

1 Machine scores as experimental outcomes

Index experimental clusters by $g = 1, \dots, G$ and observations within clusters by i . Treatment $D_g \in \{0, 1\}$ is assigned at the cluster level. Let Y_{gi} be the human-coded outcome and $\hat{Y}_{gi}$ the machine score. Define the observed scoring discrepancy

$$e_{gi} = Y_{gi} - \hat{Y}_{gi}. \quad (1)$$

The machine score must be fixed independently of the outcome analysis. A score learned on the experimental outcomes requires a separate training sample or an appropriate cross-fitting design. Swaps do not repair overfitting.

For clarity, consider the unadjusted difference in means. Write N_1 and N_0 for the numbers of observations in treated and control clusters. The proxy and gold estimates are

$$\hat{\tau}_P = \frac{1}{N_1} \sum_{g:D_g=1} \sum_i \hat{Y}_{gi} - \frac{1}{N_0} \sum_{g:D_g=0} \sum_i \hat{Y}_{gi}, \quad (2)$$

$$\hat{\tau}_G = \frac{1}{N_1} \sum_{g:D_g=1} \sum_i Y_{gi} - \frac{1}{N_0} \sum_{g:D_g=0} \sum_i Y_{gi}. \quad (3)$$

Random treatment assignment gives these contrasts their experimental interpretation. Validation sampling is a second design layered on top of the experiment. It determines what can be learned about $\hat{\tau}_G - \hat{\tau}_P$ when Y is expensive.

Define cluster g 's contribution to this endpoint gap as

$$\psi_g = \begin{cases} N_1^{-1} \sum_i e_{gi}, & D_g = 1, \\ -N_0^{-1} \sum_i e_{gi}, & D_g = 0. \end{cases} \quad (4)$$

The contributions add exactly:

$$\sum_{g=1}^G \psi_g = \hat{\tau}_G - \hat{\tau}_P. \quad (5)$$

Equation (5) is an accounting identity within the paired gold-label pool. It does not identify the errors of unlabeled clusters. For OLS with fixed covariates, residualized-treatment weights replace the treated and control weights in equation (4). The same adding-up and sampling identities follow. A nonlinear estimator can still be recomputed over swap sets, but its mean path need not be linear.

2 The fixed-budget swap distribution

Let S_k be a simple random sample without replacement of exactly k clusters. The raw-swap estimate replaces machine scores by human outcomes in the selected clusters:

$$\hat{\tau}_R(S_k) = \hat{\tau}_P + \sum_{g \in S_k} \psi_g. \quad (6)$$

The distribution of $\hat{\tau}_R(S_k)$ over all $\binom{G}{k}$ sets is the raw fixed-budget swap distribution. It describes partial repair of the observed data. A fixed k is preferable to independent Bernoulli swaps when k represents a coding budget.

Let $\bar{\psi} = G^{-1} \sum_g \psi_g$ and $S_\psi^2 = (G-1)^{-1} \sum_g (\psi_g - \bar{\psi})^2$. Standard sampling identities give

$$\mathbb{E}_{S_k}[\hat{\tau}_R(S_k)] = \hat{\tau}_P + \frac{k}{G}(\hat{\tau}_G - \hat{\tau}_P), \quad (7)$$

$$\mathbb{V}_{S_k}[\hat{\tau}_R(S_k)] = k \left(1 - \frac{k}{G}\right) S_\psi^2. \quad (8)$$

Equation (7) is the central result. In a linear outcome analysis, the average raw-swap path must be a straight line connecting the proxy and gold estimates. Any two scoring systems with the same endpoints have the same mean path. The path says nothing about whether the correction is spread evenly across clusters or concentrated in a few. Equation (8) shows where that information resides.

The distribution supports several descriptive summaries. A researcher can report its central quantiles, the chance that swapping k clusters changes the estimate's sign, and the chance that the movement exceeds a prespecified substantive threshold. These probabilities refer to the stated swap design. They are not p -values.

3 From partial swaps to a validation estimator

Raw replacement mechanically shrinks as k/G falls. If the goal is to estimate the fully human-coded contrast from a k -cluster probability sample, scale the observed correction by its inclusion probability:

$$\hat{\tau}_H(S_k) = \hat{\tau}_P + \frac{G}{k} \sum_{g \in S_k} \psi_g. \quad (9)$$

This is a Horvitz-Thompson difference correction. In this simple setting it is also the rectifier used by prediction-powered inference. It satisfies

$$\mathbb{E}_{S_k}[\hat{\tau}_H(S_k)] = \hat{\tau}_G, \quad (10)$$

$$\mathbb{V}_{S_k}[\hat{\tau}_H(S_k)] = \left(\frac{G}{k}\right)^2 k \left(1 - \frac{k}{G}\right) S_\psi^2. \quad (11)$$

Thus the design-scaled swap distribution is centered on the gold endpoint. Its spread measures validation-sample fragility at budget k .

Unequal-probability validation replaces G/k with inverse inclusion probabilities. Stratifying the

validation draw by treatment arm or by an observable proxy-risk score can improve precision. Those choices change the swap distribution and must be included in its definition. The optimal choice depends on the target estimator [1].

An exact distribution requires ψ_g for every cluster in the finite pool. That is available in a benchmark dataset, a simulation, or a large pilot used to study smaller future budgets. With only one small validation sample, the analyst cannot recover the unseen cluster contributions nonparametrically. Standard design-based variance estimation remains available when selection probabilities are known. A cluster bootstrap or a model for e_{gi} can approximate more of the distribution, but then the result depends on resampling or transport assumptions.

3.1 Why this is not the wild cluster bootstrap

The analogy to a wild cluster bootstrap is useful but incomplete. A wild cluster bootstrap applies mean-zero random multipliers to cluster score contributions to approximate a null sampling distribution [3]. A validation swap applies zero-one inclusion indicators to observed gold-minus-proxy discrepancies. Its immediate target is the effect of a labeling design, not repeated-sample uncertainty under a null.

For the linear contrast,

$$\hat{\tau}_R(S_k) - \mathbb{E}_{S_k}[\hat{\tau}_R(S_k)] = \sum_{g=1}^G \left(Z_g - \frac{k}{G} \right) \psi_g,$$

where $Z_g = \mathbf{1}\{g \in S_k\}$. This centered expression has the form of a finite-population multiplier process. The resemblance does not turn swap quantiles into confidence intervals. Valid uncertainty for the experimental effect must account for treatment assignment, and valid uncertainty for the correction must account for the actual validation design.

4 Same accuracy and endpoints, different swap distributions

The numerical example has 20 equal-size clusters with 50 observations each. Half the clusters are treated. The fully human-coded treatment effect is 0.100, and the machine-scored effect is 0.060 in the diffuse and concentrated cases. All three scoring systems have RMSE 0.200 and the same R^2 because they score the same gold outcomes and have the same squared-error total.

The error patterns differ only in their cluster contributions. In the diffuse case, each cluster contributes 0.040/20 to the endpoint gap. In the concentrated case, two clusters contribute 0.040/2 each and the others contribute zero. The offsetting case has two positive and two negative contributions of equal size, so the proxy and gold endpoints agree even though individual swaps can move the estimate.

Table 1: Matched prediction accuracy with different contribution patterns

Error pattern	Gold ATE	Proxy ATE	Gap	RMSE	R^2	Top-two share
Diffuse	0.100	0.060	0.040	0.200	0.965	10%
Concentrated	0.100	0.060	0.040	0.200	0.965	100%
Offsetting	0.100	0.100	0.000	0.200	0.965	50%

Note: Gold ATE and proxy ATE are differences in means. Gap is gold minus proxy. Top-two share is the two largest absolute cluster contributions divided by the sum of absolute contributions. Each row uses the same gold outcomes. Quantities are deterministic for the generated finite population.

Figure 1 gives the comparison the mean path misses. Panel A plots the exact mean over every fixed-size coalition. The diffuse and concentrated paths coincide. Panel B displays the exact mass function for a 5-cluster raw swap. The diffuse correction always moves by the same amount. The concentrated correction does not move 55.3% of the time because most coalitions contain neither high-error cluster.

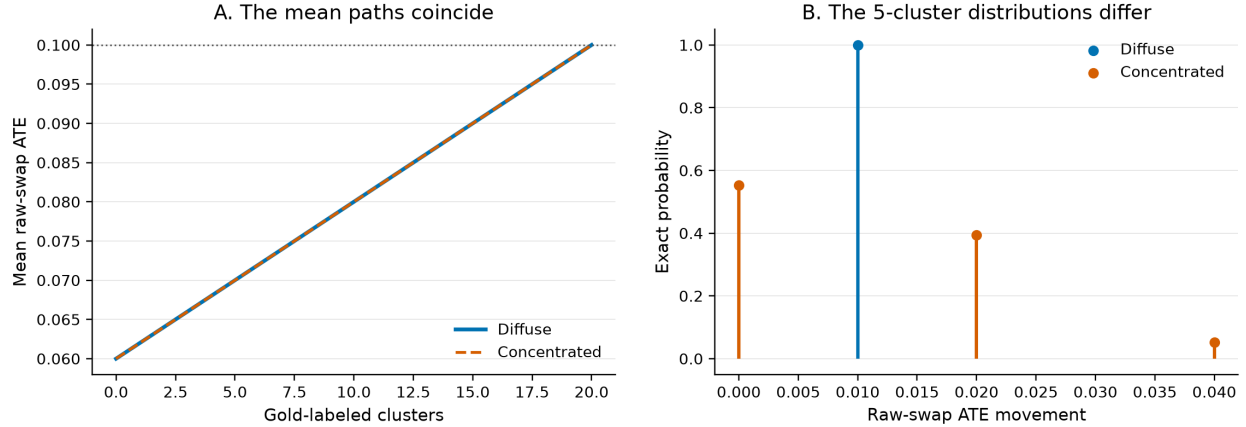


Figure 1: The mean swap path and exact fixed-budget distributions. Panel A shows the mean raw-swap ATE over every set of a given size. The horizontal dotted line is the gold ATE. Panel B shows the exact probability mass of the raw ATE movement for every 5-cluster subset. Both panels use the same 20-cluster finite population.

5 What a validation budget can deliver

Figure 2 shows the design-scaled estimator in equation (9). The line is the median over all fixed-size validation sets. Dark and light regions contain the middle 50 and 90 percent of the exact distribution. Each distribution has mean 0.100 by construction, but its shape depends on the allocation of ψ_g .

The diffuse case is stable even with one labeled cluster because the example assigns every cluster the same contribution. Concentrated error produces a discrete and uneven distribution. Offsetting error is more revealing: the proxy and gold endpoints are identical, yet small validation samples can imply large positive or negative corrections. Endpoint agreement is therefore compatible with fragile validation evidence.

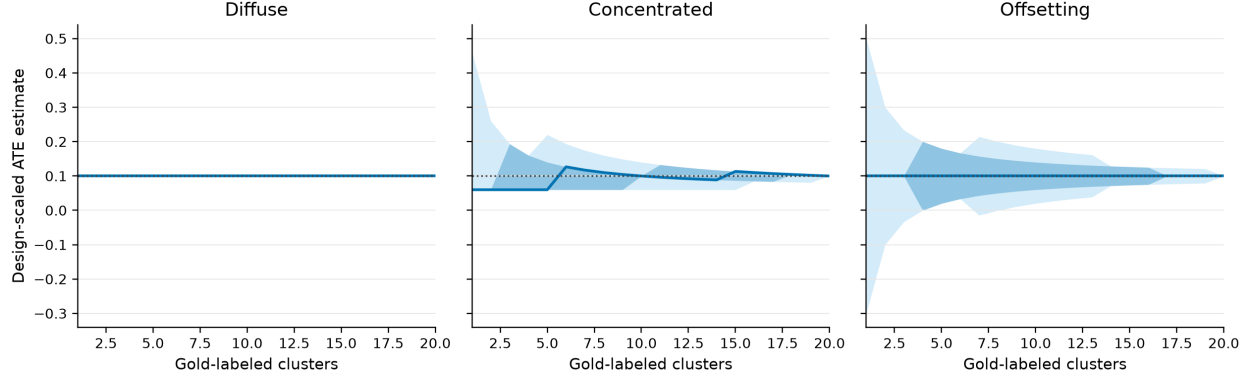


Figure 2: Exact design-scaled swap distributions by validation budget. The solid line is the median; dark and light regions contain the middle 50 and 90 percent of all fixed-size validation sets. The dotted line is the fully gold-labeled ATE. All panels share an axis.

Table 2 reports exact numerical summaries. Tail risk is the share of validation sets whose scaled estimate differs from the gold endpoint by more than 0.050. Bias is omitted because it is zero up to numerical precision for every cell. A narrow distribution does not prove that human labels measure the intended construct. It says only that the specified validation draw estimates the observed gold-label endpoint consistently.

Table 2: Exact design-scaled distributions at selected budgets

Error pattern	Gold clusters	Mean	SD	5th	95th	Tail risk
Diffuse	2	0.100	0.000	0.100	0.100	0%
	5	0.100	0.000	0.100	0.100	0%
	10	0.100	0.000	0.100	0.100	0%
	20	0.100	0.000	0.100	0.100	0%
Concentrated	2	0.100	0.083	0.060	0.260	19%
	5	0.100	0.048	0.060	0.220	5%
	10	0.100	0.028	0.060	0.140	0%
	20	0.100	0.000	0.100	0.100	0%
Offsetting	2	0.100	0.123	-0.100	0.300	35%
	5	0.100	0.071	0.020	0.180	57%
	10	0.100	0.041	0.020	0.180	14%
	20	0.100	0.000	0.100	0.100	0%

Note: Mean, SD, and quantiles describe equation (9) over all fixed-size cluster samples. Tail risk is $\Pr(|\hat{\tau}_H - \hat{\tau}_G| > 0.050)$. Enumeration is exact, not Monte Carlo. Gold ATE equals 0.100 in every scenario.

6 Use and limits

A validation-swap analysis begins with the same choices required by any two-phase design. The researcher states the causal estimand, fixes the machine scoring rule, identifies the unit sampled for human coding, and records every cluster’s inclusion probability. If treatment is assigned by cluster, swapping individual labels while treating them as independent understates the relevant variation.

The primary analysis should report a valid correction estimator and interval, using prediction-

powered, design-based supervised learning, or another method whose assumptions fit the design. The swap distribution is an additional diagnostic. Before seeing it, the researcher chooses validation budgets and a substantively meaningful movement threshold. The exhibit then reports quantiles, tail risk, and any decision reversal probability. Accuracy metrics remain useful descriptions of the scoring model, but they do not substitute for this coefficient-specific analysis.

The diagnostic has four limits. First, it cannot reveal the individual errors of unlabeled observations. Second, a distribution computed inside a convenience validation sample need not transport outside it. Third, human labels can contain their own error or define the wrong construct. Fourth, a small number of validated clusters may leave the distribution too poorly estimated to support fine subgroup claims. The proper conclusion in that case is that the validation budget does not resolve the scoring risk.

Machine-scored experiments create two sources of uncertainty: treatment assignment and gold-label sampling. Randomization-based or cluster-robust inference addresses the first. Established two-phase estimators address the second when they correct the machine-scored contrast. The random-swap distribution makes validation-sample dependence visible by showing how much the conclusion depends on which clusters were checked.

References

- [1] Gustavo Amorim, Ran Tao, Sarah Lotspeich, Pamela A. Shaw, Thomas Lumley, and Bryan E. Shepherd. 2021. Two-phase sampling designs for data validation in settings with covariate measurement error and continuous outcome. *Journal of the Royal Statistical Society: Series A* 184(4):1368-1389.
- [2] Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnica. 2023. Prediction-powered inference. *Science* 382(6671):669-674.
- [3] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. 2008. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics* 90(3):414-427.
- [4] Naoki Egami, Christian J. Fong, Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. 2022. How to make causal inferences using texts. *Science Advances* 8(42):eabg2652.
- [5] Naoki Egami, Musashi Hinck, Brandon M. Stewart, and Hanying Wei. 2023. Using imperfect surrogates for downstream inference: Design-based supervised learning for social science applications of large language models. *Advances in Neural Information Processing Systems* 36.
- [6] Siruo Wang, Tyler H. McCormick, and Jeffrey T. Leek. 2020. Methods for correcting inference based on outcomes predicted by machine learning. *Proceedings of the National Academy of Sciences* 117(48):30266-30275.